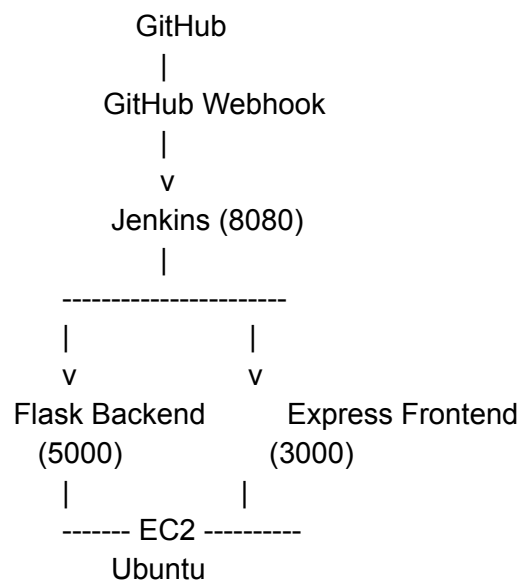


This assignment can be completed on **one Ubuntu EC2 instance** running:

- Flask Backend → Port **5000**
- Express Frontend → Port **3000**
- Jenkins → Port **8080**
- Nginx (optional reverse proxy) → Port **80**

=====

Architecture:



PART 1 – Deploy Flask and Express on EC2

Step 1: Launch EC2

Launch an Ubuntu Server 22.04 LTS EC2 instance.

Select t2.micro as the instance type (Free Tier eligible).

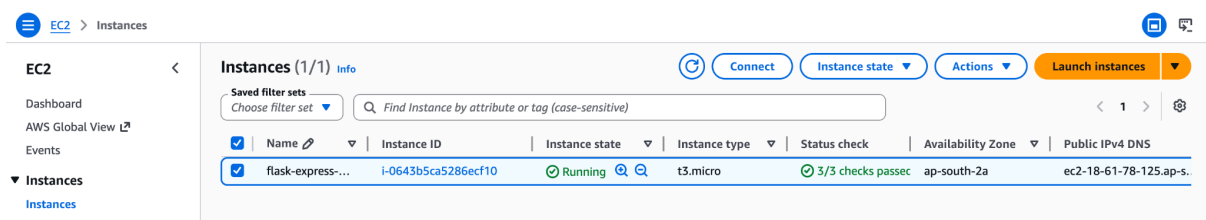
Configure 20 GB of storage.

Create or select an existing key pair for SSH access.

Configure the Security Group with the following inbound rules:

- Port 22 – SSH access
- Port 80 – Website (HTTP)
- Port 3000 – Express Frontend
- Port 5000 – Flask Backend
- Port 8080 – Jenkins Dashboard

Launch the instance and note the public IP address.



Step 2: Connect EC2

`ssh -i mykey.pem ubuntu@PUBLIC_IP`

```
rushikeshdhule@Rushikeshs-MacBook-Air ~ % ls
Applications      Downloads          iCloud Drive (Archive) - 1  mirame             Public              untitled folder
CICD_pipeline     Flask_app          kubernetess_labs          Movies
cleanup.sh        Flask-express-AWS  Library                  Music
Desktop           FlaskAppDock       miniconda3               Pictures
Documents         iCloud Drive (Archive)  minikube-darwin-arm64    Postman
rushikeshdhule@Rushikeshs-MacBook-Air ~ % cd downloads
rushikeshdhule@Rushikeshs-MacBook-Air downloads % ls
-rshl_dhule_resume_2825.docx  flask-express-app.pem  New Doc 05-17-2026 21.01.pdf  Rushikesh_Dhule_CV.pdf
AdmissionCertificate.pdf     iamuser_accessKeys.csv  New Doc 05-17-2026 21.11.pdf  Rushikesh_Dhule_Picnic.docx
animeslook.png              introduction video.mov  OnVUE.app                    Rushikesh_Dhule_Resume.pdf
AWS_Mock_Interview_QA.pdf    itadmln_accessKeys.csv  pan.jpg                      RushiPassport_280KB.jpg
aws-learning                 kingsley-zikora-devOps-Professional-Resume.pdf  Pilot_House.doc              RushiPassport.png
Caste_Certificate_under_380KB.jpg  mirame.zip              Receipt_Details_5481611.html  rutu-doc
Certificate.pdf               Motivation_Letter_Rushi.pdf  Resume                       WhatsApp Image 2026-05-05 at 17.48.27.jpeg
Deloitte                     Motivation_Letter_Rushikesh.pdf  Rushikesh_Dhule_Cover_Letter.pdf  WhatsApp Image 2026-05-05 at 17.48.51.jpeg
rushikeshdhule@Rushikeshs-MacBook-Air downloads % chmod 400 ~/downloads/flask-express-app.pem
rushikeshdhule@Rushikeshs-MacBook-Air downloads % ls -l ~/downloads/flask-express-app.pem
-rw-r--r-- 1 rushikeshdhule staff 1574 25 May 21:56 /Users/rushikeshdhule/downloads/flask-express-app.pem
rushikeshdhule@Rushikeshs-MacBook-Air downloads % ssh -i /Users/rushikeshdhule/downloads/flask-express-app.pem ubuntu@18.61.78.125
The authenticity of host '18.61.78.125 (18.61.78.125)' can't be established.
ED25519 key fingerprint is: SHA256:si8XvYv4NIUtzEnem83gv/ShewBtLVQWtn49n3ECCNU
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '18.61.78.125' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1004-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon May 25 16:33:28 UTC 2026

System load:  0.08      Temperature:   -273.1 C
Usage of /:   11.0% of 18.25GB    Processes:    121
Memory usage: 26%              Users logged in: 0
Swap usage:   0%               IPv4 address for ens5: 172.31.12.108

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-12-188:~$
```

Step 3: Update Server

```
sudo apt update  
sudo apt upgrade -y
```

Step 4: Install Git

```
sudo apt install git -y
```

Step 5: Install Python

```
sudo apt install python3 python3-pip python3-venv -y
```

Step 6: Install NodeJS

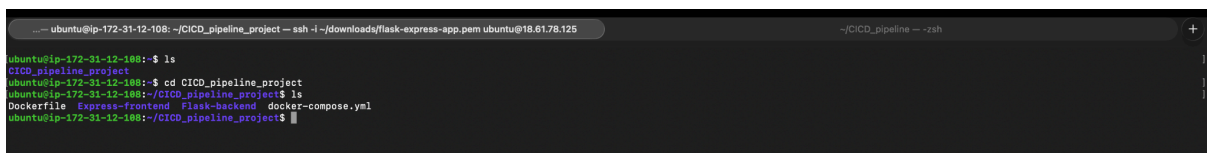
```
curl -fsSL https://deb.nodesource.com/setup_18.x | sudo -E bash -  
sudo apt install nodejs -y
```

Step 7: Install PM2

```
sudo npm install -g pm2
```

Step 8: Clone Repositories

```
Git clone https://github.com/21Rushi/CICD_pipeline_project.git
```



```
-- ubuntu@ip-172-31-12-108: ~/CICD_pipeline_project -- ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125 --/CICD_pipeline -- -zsh  
ubuntu@ip-172-31-12-108:~$ ls  
CICD_pipeline_project  
ubuntu@ip-172-31-12-108:~$ cd CICD_pipeline_project  
ubuntu@ip-172-31-12-108:~/CICD_pipeline_project$ ls  
Dockerfile Express-frontend Flask-backend docker-compose.yml  
ubuntu@ip-172-31-12-108:~/CICD_pipeline_project$
```

Step 9: Deploy Flask Backend

```
cd flask-backend  
python3 -m venv venv  
source venv/bin/activate  
pip install -r requirements.txt
```

Test→ python [app.py](#)

<http://18.61.78.125:5000/>

```

[ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project$ cd Flask-backend
[ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ python3 -m venv venv
[ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ source venv/bin/activate
(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ pip install -r requirements.txt
Collecting bcrypt==5.0.0 (from -r requirements.txt (line 1))
  Downloading bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl.metadata (10 kB)
Collecting blinker==1.9.0 (from -r requirements.txt (line 2))
  Downloading blinker-1.9.0-py3-none-any.whl.metadata (1.6 kB)
Collecting click==8.3.1 (from -r requirements.txt (line 3))
  Downloading click-8.3.1-py3-none-any.whl.metadata (2.6 kB)
Collecting dnspython==1.16.0 (from -r requirements.txt (line 4))
  Downloading dnspython-1.16.0-py2.py3-none-any.whl.metadata (1.8 kB)
Collecting Flask==3.1.2 (from -r requirements.txt (line 5))
  Downloading flask-3.1.2-py3-none-any.whl.metadata (3.2 kB)
Collecting itsdangerous==2.2.0 (from -r requirements.txt (line 6))
  Downloading itsdangerous-2.2.0-py3-none-any.whl.metadata (1.9 kB)
Collecting Jinja2==3.1.6 (from -r requirements.txt (line 7))
  Downloading Jinja2-3.1.6-py3-none-any.whl.metadata (2.9 kB)
Collecting MarkupSafe==3.0.3 (from -r requirements.txt (line 8))
  Downloading markupsafe-3.0.3-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl.metadata (2.7 kB)
Collecting pymongo==3.12.0 (from -r requirements.txt (line 9))
  Downloading pymongo-3.12.0.tar.gz (818 kB)
    ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 818.6/818.6 kB 41.9 MB/s eta 0:00:00
Installing build dependencies ... done
Getting requirements to build wheel ... done
Preparing metadata (pyproject.toml) ... done
Collecting python-dotenv==1.2.1 (from -r requirements.txt (line 10))
  Downloading python_dotenv-1.2.1-py3-none-any.whl.metadata (25 kB)
Collecting Werkzeug==3.1.5 (from -r requirements.txt (line 11))
  Downloading werkzeug-3.1.5-py3-none-any.whl.metadata (4.0 kB)
Downloaded bcrypt-5.0.0-cp39-abi3-manylinux_2_34_x86_64.whl (278 kB)
Downloaded blinker-1.9.0-py3-none-any.whl (10.5 kB)
Downloaded click-8.3.1-py3-none-any.whl (108 kB)
Downloaded dnspython-1.16.0-py2.py3-none-any.whl (188 kB)
Downloaded flask-3.1.2-py3-none-any.whl (103 kB)
Downloaded itsdangerous-2.2.0-py3-none-any.whl (16 kB)
Downloaded Jinja2-3.1.6-py3-none-any.whl (134 kB)
Downloaded markupsafe-3.0.3-cp314-cp314-manylinux2014_x86_64.manylinux_2_17_x86_64.manylinux_2_28_x86_64.whl (23 kB)
Downloaded python_dotenv-1.2.1-py3-none-any.whl (21 kB)
Downloaded werkzeug-3.1.5-py3-none-any.whl (225 kB)
Building wheels for collected packages: pymongo
  Building wheel for pymongo (pyproject.toml) ... done
  Created wheel for pymongo: filename=pymongo-3.12.0-cp314-linux_x86_64.whl size=515115 sha256=95183ec9a45b2f46465621ae6ef58a2f11acf2744b5d4a75b59ff823e1ca2c9b
  Stored in directory: /home/ubuntu/.cache/pip/wheels/93/2f/4a/49e8864c2c53cdc281882377d2a6df299749b79d3b3de2aae6
Successfully built pymongo
Installing collected packages: python-dotenv, pymongo, MarkupSafe, itsdangerous, dnspython, click, blinker, bcrypt, Werkzeug, Jinja2, Flask
Successfully installed Flask-3.1.2 Jinja2-3.1.6 MarkupSafe-3.0.3 Werkzeug-3.1.5 bcrypt-5.0.0 blinker-1.9.0 click-8.3.1 dnspython-1.16.0 itsdangerous-2.2.0 pymongo-3.12.0 python-dotenv-1.2.1
(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ ]

```

```

...172-31-12-108:~/CI/CD_pipeline_project/Flask-backend - ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125
(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ python app.py
Mongo URI: mongodb+srv://FlaskApp:welcome123@flaskapldb.m2j9no9.mongodb.net/?appName=FlaskApiDBnmnp
Mongo Connected DBs: ['FlaskDB', 'admin', 'local']
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://172.31.12.108:5000
* Running on http://172.31.12.108:5000
Press CTRL+C to quit
* Restarting with stat
Mongo URI: mongodb+srv://FlaskApp:welcome123@flaskapldb.m2j9no9.mongodb.net/?appName=FlaskApiDBnmnp
Mongo Connected DBs: ['FlaskDB', 'admin', 'local']
* Debugger is active!
* Debugger PIN: 616-289-872

```

18.61.78.125:5000

Not Secure

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Welcome to Flask Mongo API

Step 10: Run Flask Using PM2

```

pm2 start app.py \
--interpreter python3 \
--name flask-backend

```

Check → pm2 list

Save → pm2 save

```

...172-31-12-108:~/CI/CD_pipeline_project/Flask-backend - ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125
(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ pm2 start app.py \
--interpreter python3 \
--name flask-backend
[PM2] Starting /home/ubuntu/CI/CD_pipeline_project/Flask-backend/app.py in fork_mode (1 instance)
[PM2] Done.

```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	flask-backend	default	N/A	fork	13547	0s	0	online	0%	10.2mb	ubuntu	disabled

```

(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ pm2 list

```

id	name	namespace	version	mode	pid	uptime	o	status	cpu	mem	user	watching
0	flask-backend	default	N/A	fork	13547	10s	0	online	0%	45.5mb	ubuntu	disabled

```

(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
(venv) [ubuntu@ip-172-31-12-108:~/CI/CD_pipeline_project/Flask-backend$ ]

```

Step 11: Deploy Express Frontend

```
cd ~/projects/express-frontend
```

```
npm install
```

Start → node [server.js](#)

```
...31-12-108: ~/CICD_pipeline_project/Express-frontend -- ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125
~/CICD_pipeline -- -zsh

(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project$ cd Express-frontend
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ npm install

up to date, audited 86 packages in 2s

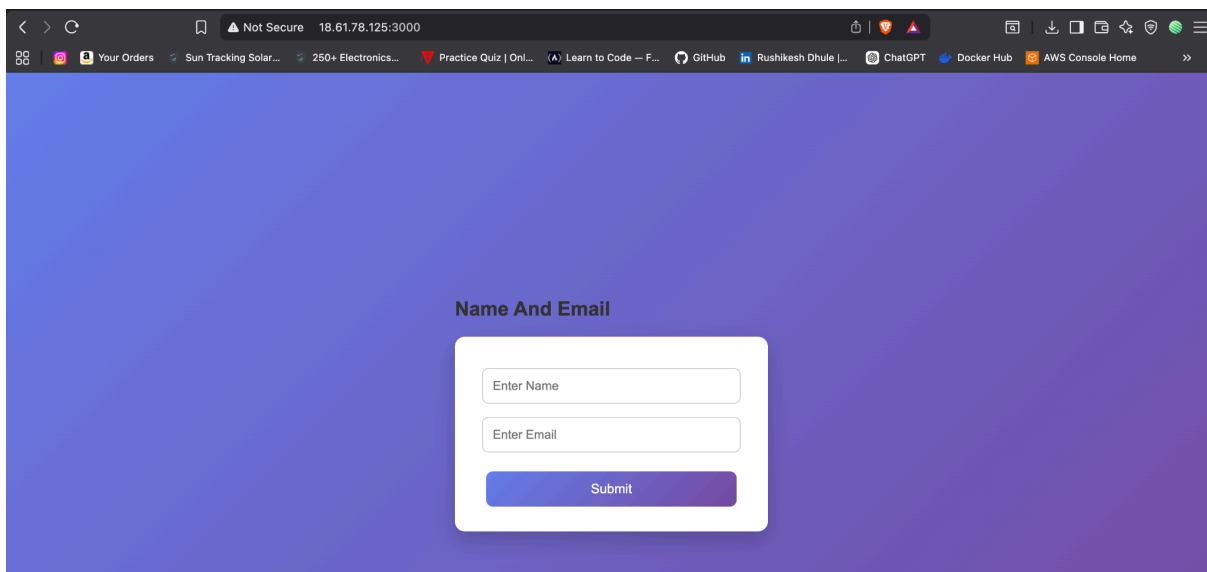
17 packages are looking for funding
  run `npm fund` for details

0 vulnerabilities (5 moderate, 3 high)

To address all issues, run:
  npm audit fix

Run `npm audit` for details.
npm notice New major version of npm available! 10.8.2 -> 11.15.0
npm notice Changelog: https://github.com/npm/cli/releases/tag/v11.15.0
npm notice To update run: npm install -g npm@11.15.0
npm notice
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ node server.js
Frontend running on port 3000
```

Check → <http://18.61.78.125:3000/>



Step 12: Run Express Using PM2

```
pm2 start server.js \
```

```
--name express-frontend
```

Check → pm2 list

Save → pm2 save

Enable startup → pm2 startup

Save → pm2 save

```
...31-12-108: ~/CICD_pipeline_project/Express-frontend -- ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125 ~|CICD_pipeline -- -zsh
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ pm2 start server.js \
--name express-frontend
[PM2] Starting /home/ubuntu/CICD_pipeline_project/Express-frontend/server.js in fork_mode (1 instance)
[PM2] Done.

```

id	name	namespace	version	mode	pid	uptime	⚡	status	cpu	mem	user	watching
1	express-frontend	default	1.0.0	fork	13714	0s	0	online	0%	15.6mb	ubuntu	disabled
0	flask-backend	default	N/A	fork	13547	3m	0	online	0%	45.6mb	ubuntu	disabled

```
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ pm2 list

```

id	name	namespace	version	mode	pid	uptime	⚡	status	cpu	mem	user	watching
1	express-frontend	default	1.0.0	fork	13714	14s	0	online	0%	56.7mb	ubuntu	disabled
0	flask-backend	default	N/A	fork	13547	3m	0	online	0%	45.6mb	ubuntu	disabled

```
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ pm2 startup
[PM2] Init System found: systemd
[PM2] To setup the Startup Script, copy/paste the following command:
sudo env PATH=$PATH:/usr/bin /usr/lib/node_modules/pm2/bin/pm2 startup systemd -u ubuntu --hp /home/ubuntu
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$ pm2 save
[PM2] Saving current process list...
[PM2] Successfully saved in /home/ubuntu/.pm2/dump.pm2
(venv) ubuntu@ip-172-31-12-108:~/CICD_pipeline_project/Express-frontend$
```

PART 2 – Install Jenkins

Step 13: Install Java

```
sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

```
sudo apt update
sudo apt install fontconfig openjdk-21-jre
java -version
```

Step 14: Install Jenkins

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/" | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
sudo apt update
sudo apt install jenkins
```

Start →

```
sudo systemctl enable jenkins
sudo systemctl start jenkins
```



```
--/downloads -- ubuntu@ip-172-31-12-108: ~ -- ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125 --/C

(env) ubuntu@ip-172-31-12-108:~$ sudo apt update
sudo apt install jenkins -y
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:2 https://pkg.jenkins.io/debian-stable binary/ Release
Hit:4 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:5 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:6 http://ap-south-2.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:7 https://deb.nodesource.com/node_18.x nodistro InRelease
Hit:8 http://security.ubuntu.com/ubuntu resolute-security InRelease
All packages are up to date.
jenkins is already the newest version (2.555.2).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 0
(env) ubuntu@ip-172-31-12-108:~$
```

Verify status : `sudo systemctl status jenkins`

```
--/downloads -- ubuntu@ip-172-31-12-108: ~ -- ssh -i ~/downloads/flask-express-app.pem ubuntu@18.61.78.125 --/CICD_pipeline -- -zsh

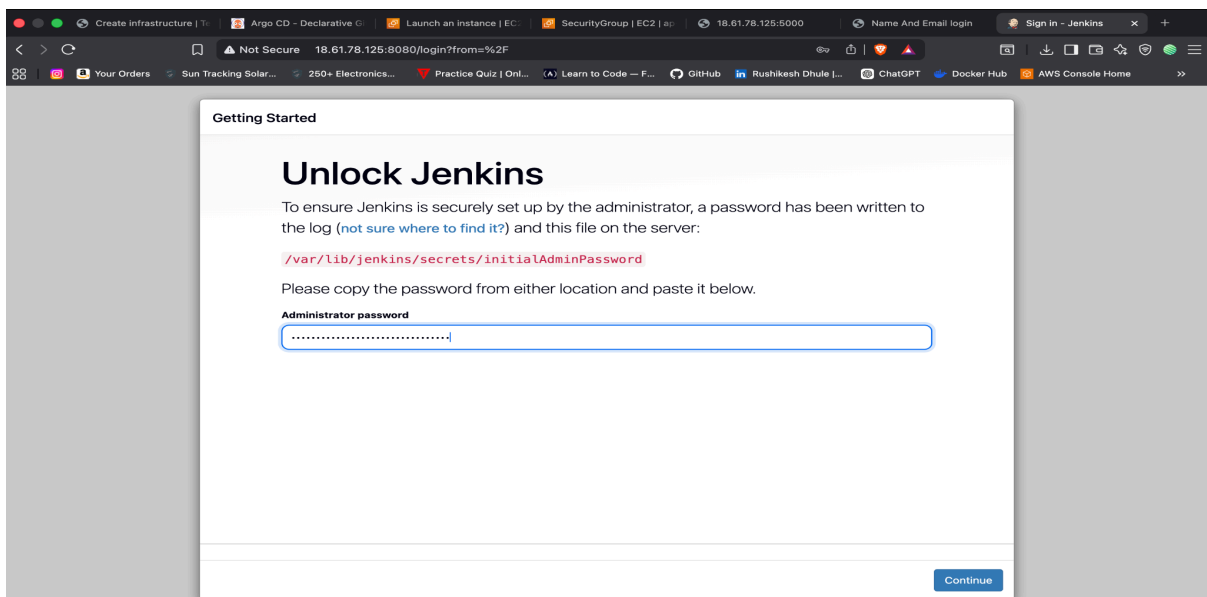
(env) ubuntu@ip-172-31-12-108:~$ sudo systemctl enable jenkins
sudo systemctl start jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
(env) ubuntu@ip-172-31-12-108:~$ sudo systemctl status jenkins
jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Mon 2026-05-25 17:19:23 UTC; 1min 31s ago
     Invocation: 31607494be7e42a7bb7899bc9aee748e
   Main PID: 16493 (java)
      Tasks: 37 (limit: 627)
     Memory: 310M (peak: 323M)
        CPU: 20.920s
   CGroup: /system.slice/jenkins.service
           └─16493 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

May 25 17:19:16 ip-172-31-12-108 jenkins[16493]: [LF]>
May 25 17:19:16 ip-172-31-12-108 jenkins[16493]: [LF]> *****
May 25 17:19:16 ip-172-31-12-108 jenkins[16493]: [LF]> *****
May 25 17:19:16 ip-172-31-12-108 jenkins[16493]: [LF]> *****
May 25 17:19:23 ip-172-31-12-108 jenkins[16493]: 2026-05-25 17:19:23.172+0000 [id=31] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
May 25 17:19:23 ip-172-31-12-108 jenkins[16493]: 2026-05-25 17:19:23.239+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
May 25 17:19:23 ip-172-31-12-108 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
May 25 17:19:24 ip-172-31-12-108 jenkins[16493]: 2026-05-25 17:19:24.998+0000 [id=50] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenIn
May 25 17:19:24 ip-172-31-12-108 jenkins[16493]: 2026-05-25 17:19:24.998+0000 [id=50] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
May 25 17:19:28 ip-172-31-12-108 jenkins[16493]: 2026-05-25 17:19:28.348+0000 [id=78] WARNING h.n.DiskSpaceMonitorDescriptor#markNodeOfflineOrOnline: Making Built-In Node offline temporarily
lines 1-21/21 (END)
```

Step 15: Get Jenkins Password

`sudo cat /var/lib/jenkins/secrets/initialAdminPassword`

Check → <http://18.61.78.125:8080/>



Step 16: Install Jenkins Plugins

Dashboard → Manage Jenkins → Plugins

Install:

- Git Plugin
- Pipeline Plugin
- NodeJS Plugin
- SSH Agent Plugin

Restart Jenkins.

Step 17: Configure Tools

Manage Jenkins → Tools

NodeJS:

NodeJS18

Install automatically.

Python already exists on server.

PART 3 – CI/CD Pipeline :

Jenkins Script →

```
pipeline {
  agent any

  stages {

    stage('Clone Repository') {
      steps {
        git branch: 'main',
        url: 'https://github.com/21Rushi/CI_CD_pipeline_project.git'
      }
    }

    stage('Install Backend Dependencies') {
```



```

steps {
  dir('Flask-backend') {
    sh ""
    pip3 install --break-system-packages -r requirements.txt
    ""
  }
}

```

```

stage('Install Frontend Dependencies') {
  steps {
    dir('Express-frontend') {
      sh ""
      npm install
      ""
    }
  }
}

```

```

stage('Deploy Backend') {
  steps {
    dir('Flask-backend') {
      sh ""
      pm2 restart flask-backend || \
      pm2 start app.py \
      --interpreter python3 \
      --name flask-backend
      ""
    }
  }
}

```

```

stage('Deploy Frontend') {
  steps {
    dir('Express-frontend') {
      sh ""
      pm2 restart express-frontend || \
      pm2 start npm \
      --name express-frontend \
      -- start
      ""
    }
  }
}

```

```

post {
  success {

```

```
        echo 'Deployment Successful'
    }

    failure {
        echo 'Deployment Failed'
    }
}
}
```

Step 18: Configure GitHub Webhook

GitHub Repository →

Settings →

Webhooks →

Add webhook

Payload URL:

`http://PUBLIC_IP:8080/github-webhook/`

Content type:

`application/json`

Events:

Just the push event

Save.

Step 19: Enable GitHub Trigger in Jenkins

Pipeline →

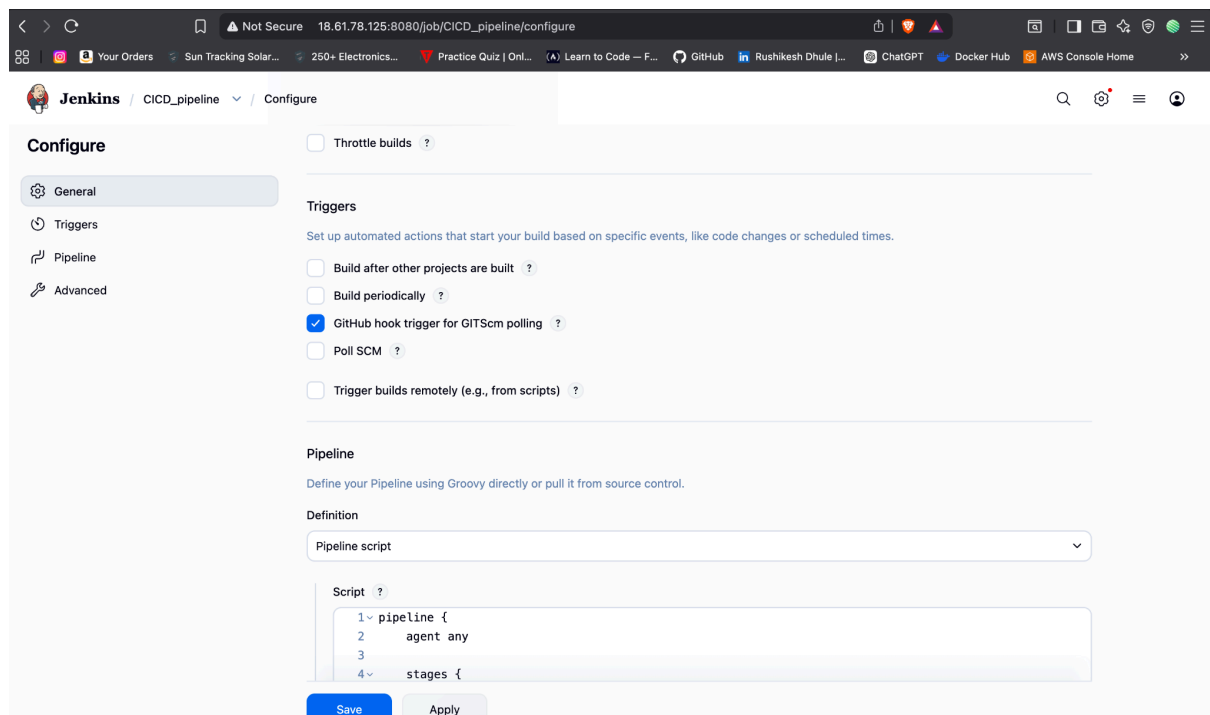
Configure →

Build Triggers →

Enable:

GitHub hook trigger for GITScm polling

Save.



Step 20: Test CI/CD

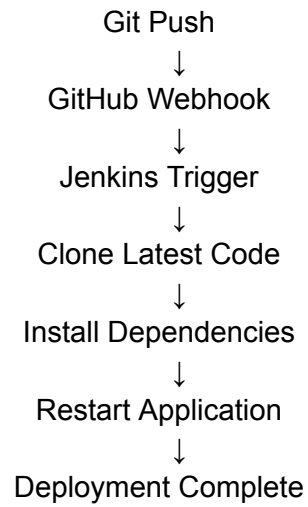
Push code:

git add .

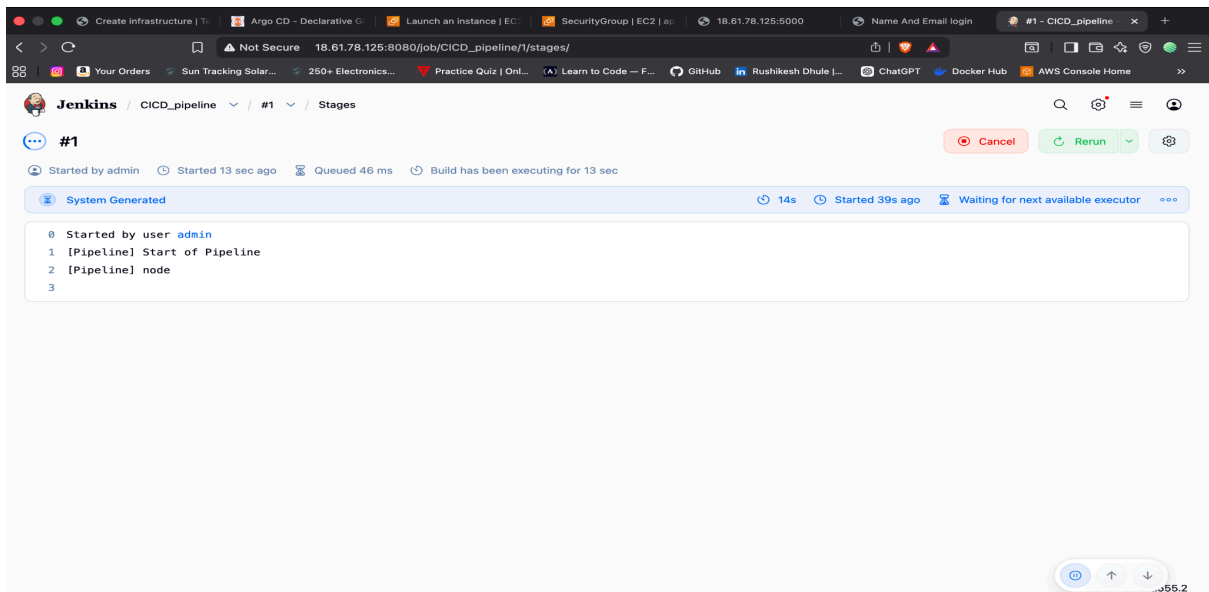
git commit -m "testing ci"

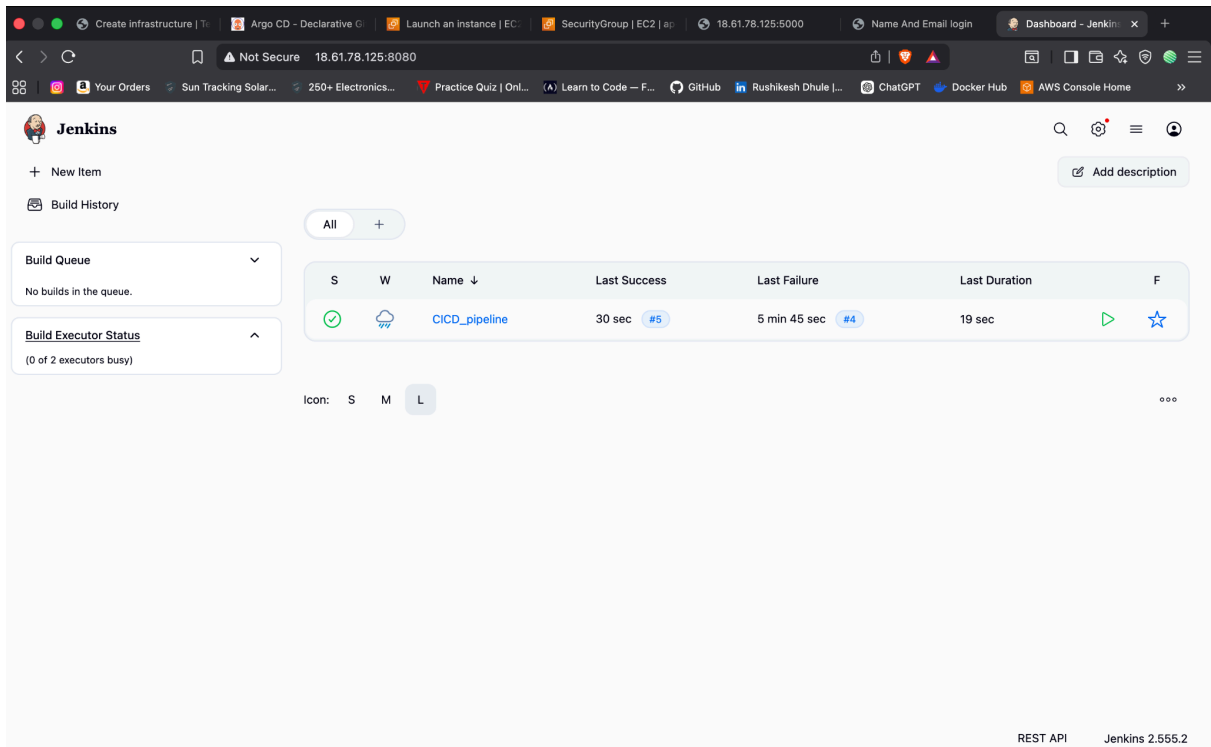
git push origin main

Flow →



Reviewed application logs, performed troubleshooting, identified and resolved build issues, and ensured the CI/CD pipeline completed successfully.



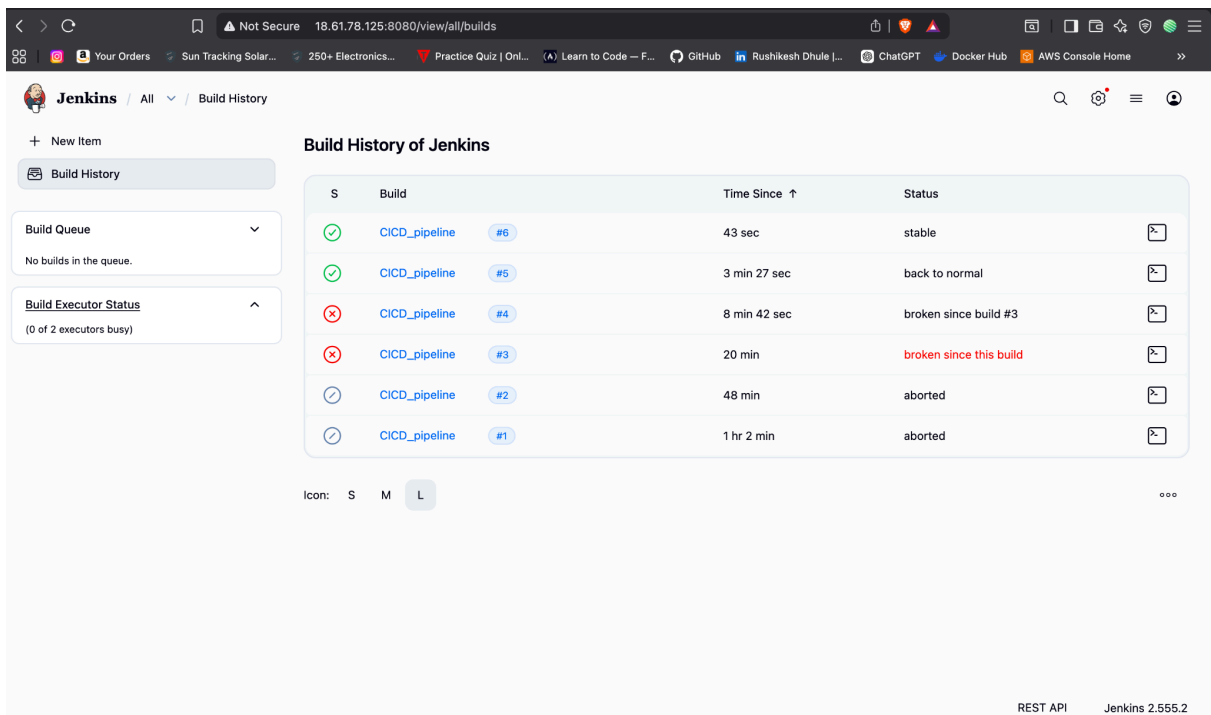


The screenshot shows the Jenkins Dashboard. The top navigation bar includes links for 'New Item', 'Build History', and 'Add description'. The main content area displays a table of build history for the 'CICD_pipeline'.

S	W	Name ↓	Last Success	Last Failure	Last Duration	F
✓	☁	CICD_pipeline	30 sec #5	5 min 45 sec #4	19 sec	▶ ☆

Icon: S M L

REST API Jenkins 2.555.2



The screenshot shows the 'Build History of Jenkins' page. The table lists the build history for the 'CICD_pipeline'.

S	Build	Time Since ↑	Status
✓	CICD_pipeline #6	43 sec	stable
✓	CICD_pipeline #5	3 min 27 sec	back to normal
✗	CICD_pipeline #4	8 min 42 sec	broken since build #3
✗	CICD_pipeline #3	20 min	broken since this build
⌚	CICD_pipeline #2	48 min	aborted
⌚	CICD_pipeline #1	1 hr 2 min	aborted

Icon: S M L

REST API Jenkins 2.555.2

Github link : https://github.com/21Rushi/CICD_pipeline_project

Google link :

https://docs.google.com/document/d/1WzWsH_RI1tVyfw1xaNG5Oo58hhMQgLyEbMeF3Dk6JaM/edit?tab=t.wrzm7vyus06